

## AI 500 Assignment - I

### Problem 1 :

$$a = 3d^2 + 2c + b^2$$

$$b = 2c^2$$

$$z = 3a + 2b^2$$

$$\therefore c = 2, d = 5$$

### 1). Forward Pass :

$$b = 2c^2$$

$$= 2(2)^2$$

$$\boxed{b = 8}$$

$$a = 3d^2 + 2c + b^2$$

$$= 3(5)^2 + 2(2) + (8)^2$$

$$\boxed{a = 443}$$

$$z = 3a + 2b^2$$

$$= 3(443) + 2(64)$$

$$\boxed{z = 1457}$$

### 2). $dz/dc$ :

paths from  $c \rightarrow z$  :

1).  $c \rightarrow a \rightarrow z$

2).  $c \rightarrow b \rightarrow a \rightarrow z$

3).  $c \rightarrow b \rightarrow z$

So,

$$\frac{dz}{dc} = \frac{\partial z}{\partial a} \cdot \frac{\partial a}{\partial c} + \frac{\partial z}{\partial a} \cdot \frac{\partial a}{\partial b} \cdot \frac{\partial b}{\partial c} + \frac{\partial z}{\partial b} \cdot \frac{\partial b}{\partial c}$$

$$\frac{\partial z}{\partial a} = \frac{\partial (3a + 2b^2)}{\partial a} = 3$$

$$\frac{\partial a}{\partial c} = \frac{\partial (3d^3 + 2c + b^2)}{\partial c} = 2$$

$$\frac{\partial a}{\partial b} = \frac{\partial (3d^3 + 2c + b^2)}{\partial b} = 2b$$

$$\frac{\partial b}{\partial c} = \frac{\partial (2c^2)}{\partial c} = 4c$$

$$\frac{\partial z}{\partial b} = \frac{\partial (3a^2 + 2b^2)}{\partial b} = 4b$$

Now,  $\frac{dz}{dc} = (3)(2) + (3)(2b)(4c) + (4b)(4c)$

$$= 6 + 24bc + 16bc$$

$$= 6 + 40bc$$

So,

$$\frac{dz}{dc} = 6 + 40(8)(2)$$

$$\therefore b = 8$$

$$\therefore c = 2$$

$$= \boxed{646}$$

## Problem 2:

### Part A Forward Pass

#### 1). First Hidden Layer:

Node  $a_{11}$ :

$$\begin{aligned} z_{11} &= x_1 w_{11} + x_2 w_{21} \\ &= (0.5)(0.9) + (0.3)(-0.79) \\ &= 0.45 + 0.237 \end{aligned}$$

$$z_{11} = 0.213$$

$$\begin{aligned} \tanh a_{11} &= \tanh(x_1 w_{11} + x_2 w_{21}) \\ &= \tanh(0.213) \end{aligned}$$

$$a_{11} = 0.2098$$

Node  $a_{12}$ :

$$\begin{aligned} z_{12} &= x_1 w_{12} + x_2 w_{22} \\ &= (0.5)(0.75) + (0.3)(-0.03) \end{aligned}$$

$$z_{12} = 0.366$$

$$\begin{aligned} a_{12} &= \tanh(x_1 w_{12} + x_2 w_{22}) \\ &= \tanh(0.366) \\ &= 0.3505 \end{aligned}$$



Node  $a_{13}$   $[1]$   $[1]$

$$z_{13} = x_1 w_{13} + x_2 w_{23}$$

$$z_{13} = (0.5)(-0.15) + (0.3)(0.1)$$

$$= -0.075 + 0.03$$

$$z_{13} = -0.045$$

$$a_{13} = \tanh(x_1 w_{13} + x_2 w_{23})$$

$$= \tanh(-0.045)$$

$$a_{13} = -0.0450$$

Node  $a_{14}$ :

$$z_{14} = x_1 w_{14} + x_2 w_{24}$$

$$= (0.5)(0.33) + (0.3)(0.02)$$

$$z_{14} = 0.171$$

$$a_{14} = \tanh(x_1 w_{14} + x_2 w_{24})$$

$$= \tanh(0.171)$$

$$a_{14} = 0.1695$$

2). Second Hidden Layer:

Node  $a_{21}$ :

$$z_{21} = a_{11} w_{11} + a_{12} w_{21} + a_{13} w_{31} + a_{14} w_{41}$$

$$= (0.2098)(-0.58) + (0.3505)(0.23)$$

$$+ (-0.0450)(0.25) + (0.1695)(-0.57)$$

$$= -0.1217 + 0.0806 - 0.01125 - 0.0966$$

$$Z_{21} = -0.1489$$

$$a_{21} = \tanh(z_{21})$$

$$= \tanh(-0.1489)$$

$$a_{21} = -0.1479$$

Node  $a_{22}$ :

$$Z_{22} = a_{11}^{[2]} w_{12}^{[2]} + a_{12}^{[2]} w_{22}^{[2]} + a_{13}^{[2]} w_{32}^{[2]} + a_{14}^{[2]} w_{42}^{[2]}$$

$$= (0.2098)(1.0) + (0.3505)(0.45) + (-0.0450)(-0.55) + (0.1695)(0.51)$$

$$= 0.2098 + 0.1577 + 0.02425 + 0.0864$$

$$Z_{22} = 0.4832$$

$$a_{22} = \tanh(a_{11}^{[2]} w_{12}^{[2]} + a_{12}^{[2]} w_{22}^{[2]} + a_{13}^{[2]} w_{32}^{[2]} + a_{14}^{[2]} w_{42}^{[2]})$$

$$= \tanh(0.483)$$

$$a_{22} = 0.4489$$

Node  $a_{23}$ :

$$Z_{23} = a_{11}^{[2]} w_{13}^{[2]} + a_{12}^{[2]} w_{23}^{[2]} + a_{13}^{[2]} w_{33}^{[2]} + a_{14}^{[2]} w_{43}^{[2]}$$

$$= (0.2098)(-0.8) + (0.3505)(0.05) + (-0.0450)(-0.09) + (0.1695)(-0.33)$$



$$Z_{23} = -0.2022$$

$$a_{23} = \tanh(Z_{23})$$

$$= \tanh(-0.2022)$$

$$a_{23} = -0.1995$$

3). Output Layer

output  $o_1$

$$Z_{o_1} = a_{21} w_{11}^{[3]} + a_{22} w_{21}^{[3]} + a_{23} w_{31}^{[3]}$$

$$= (-0.1479)(0.97) + (0.4489)(-0.01)$$

$$+ (-0.1995)(0.85)$$

$$Z_{o_1} = -0.3175$$

output  $o_2$

$$Z_{o_2} = a_{21} w_{12}^{[3]} + a_{22} w_{22}^{[3]} + a_{23} w_{32}^{[3]}$$

$$= (-0.1479)(0.97) + (0.4489)(-0.2)$$

$$+ (-0.1995)(0.76)$$

$$Z_{o_2} = -0.2858$$

## Part B Backward <sup>pa</sup> Propagation & weight updates

Setting,

Binary cross-entropy loss for one output.

$$L = -[y \ln(o) + (1-y) \ln(1-o)]$$

$$\frac{\partial L}{\partial z_0} = \frac{\partial L}{\partial o} \cdot \frac{\partial o}{\partial z_0}$$

$$= o - y$$

$$\therefore \delta_{o_1} = o_1 - y_1 = 0.4213 - 0.4$$

$$\boxed{\delta_{o_1} = 0.0213}$$

$$\therefore \delta_{o_2} = o_2 - y_2 = 0.4290 - 0.45$$

$$\boxed{\delta_{o_2} = -0.0210}$$

these represents  $\partial E / \partial z_{o_1}$  and  $\partial E / \partial z_{o_2}$  directly. ( $E = \text{total loss} = L_1 + L_2$ )

$$1). \frac{\partial E}{\partial w_{31}} [3]$$

The weight connects  $q_{23} \rightarrow o_1$ .



$$\frac{\partial E}{\partial w_{31}^{[3]}} = \frac{\partial E}{\partial z_{o1}} \cdot \frac{\partial z_{o1}}{\partial w_{31}^{[3]}}$$

Since,

$$\therefore z_{o1} = q_{21}w_{11}^{[3]} + q_{22}w_{21}^{[3]} + q_{23}w_{31}^{[3]}$$

$$\Rightarrow \frac{\partial z_{o1}}{\partial w_{31}^{[3]}} = q_{23}$$

$$\frac{\partial E}{\partial w_{31}^{[3]}} = \delta_{o1} \times q_{23}$$

$$= (0.0213)(-0.1995)$$

$$\boxed{\frac{\partial E}{\partial w_{31}^{[3]}} = -0.00425}$$

$$2). \frac{\partial E}{\partial w_{23}^{[2]}}$$

The weight connects  $q_{12} \rightarrow q_{23}$

$$\frac{\partial E}{\partial q_{23}} = \delta_{o1} \cdot w_{31}^{[3]} + \delta_{o2} \cdot w_{32}^{[3]} \quad \hookrightarrow \text{hidden node}$$

$$= (0.0213)(0.85) + (-0.0210)(0.76)$$

$$= 0.018105 - 0.01596$$

$$= 0.002145$$

$\Rightarrow$  how much  $q_{23}$  affect the total error



Now,

converting this into error at  $z_{23}$ , using tanh derivative

$$\begin{aligned}\tanh'(x) &= 1 - \tanh^2(x) \\ &= 1 - a_{23}^2 \\ &= 1 - (0.1995)^2 \\ &= 0.9602\end{aligned}$$

$$\begin{aligned}\delta a_{23} &= \frac{\partial E}{\partial z_{23}} = \frac{\partial E}{\partial a_{23}} (1 - a_{23}^2) \\ &= (0.002145)(0.9602) \\ &= 0.002060\end{aligned}$$

Now,

multiplying by the input, which is  $a_{12}$

Since,

$$\therefore z_{23} = \dots + a_{12} w_{23}^{[2]} + \dots$$

$$\begin{aligned}\frac{\partial E}{\partial w_{23}^{(2)}} &= \delta_{23} \times a_{12} \\ &= (0.002060)(0.3505)\end{aligned}$$

$$\boxed{\frac{\partial E}{\partial w_{23}^{(2)}} = 0.000722}$$

$$3). \frac{\delta E}{\delta w_{12}}$$

This weight connects  $x_1 \rightarrow a_{12}$

$$\begin{aligned} \frac{\delta E}{\delta a_{21}} &= \delta o_1 (0.97) + \delta o_2 (0.3) \\ &= (0.0213)(0.97) + (-0.0210)(0.3) \\ &= 0.014361 \end{aligned}$$

$$\begin{aligned} \frac{\delta E}{\delta a_{22}} &= \delta o_1 (-0.01) + \delta o_2 (-0.2) \\ &= (0.0213)(-0.01) + (-0.0210)(-0.2) \\ &= 0.003987 \end{aligned}$$

So, Now,

$$\begin{aligned} 1 - a_{21}^2 &= 1 - (0.1479)^2 = 1 - 0.02187 \\ &= 0.9781 \end{aligned}$$

$$\Rightarrow \delta_{21} = 0.014361 \times 0.9781$$

$$\boxed{\delta_{21} = 0.014048}$$

$$\begin{aligned} 1 - a_{22}^2 &= 1 - (0.4489)^2 = 1 - 0.2015 \\ &= 0.7985 \end{aligned}$$

$$\Rightarrow \delta_{22} = 0.003987 \times 0.7985$$

$$\boxed{\delta_{22} = 0.003184}$$



$$1 - q_{23}^2 = 0.9602$$

$$\Rightarrow \delta q_3 = 0.002145 \times 0.9602$$

$$\boxed{\delta q_{23} = 0.002060}$$

Now,

$$\frac{\partial E}{\partial q_{12}} = \delta_{21} w_{21}^{[2]} + \delta_{22} w_{22}^{[2]} + \delta_{23} w_{23}^{[2]}$$

$$= (0.014048)(0.23) +$$

$$(0.003184)(0.45) +$$

$$(0.002060)(0.05)$$

$$\boxed{\frac{\partial E}{\partial q_{12}} = 0.004767}$$

$$1 - q_{12}^2 = 1 - (0.3505)^2$$

$$= 1 - 0.1229$$

$$= 0.8772$$

$$\delta_{12} = 0.004767 \times 0.8772$$

$$= 0.004182$$

$$\frac{\partial E}{\partial w_2^{[1]}} = \delta_{12} \times x_1 = (0.004182)(0.5)$$

$$\frac{\delta E}{\delta w_{12}^{(1)}} = 0.002091$$

#### 4). Weights Updated

$$w_{new} = w_{old} - \eta \frac{\delta E}{\delta w}, \quad \eta = 0.01$$

$$w_{31}^{(3)} :$$

$$\begin{aligned} w_{31}^{(3), new} &= 0.85 - 0.01(-0.00425) \\ &= 0.85 - 0.0000425 \\ &= \boxed{0.8500425} \end{aligned}$$

$$w_{23}^{(2)} :$$

$$\begin{aligned} w_{23}^{(2), new} &= 0.05 - 0.01(0.000722) \\ &= 0.05 - 0.00000722 \\ &= \boxed{0.0499928} \end{aligned}$$

$$w_{12}^{(1)} :$$

$$\begin{aligned} w_{12}^{(1), new} &= 0.75 - 0.01(0.002091) \\ &= 0.75 - 0.00002091 \\ &= \boxed{0.7499791} \end{aligned}$$